Sri Lanka Institute of Information Technology



Green DevOps - Energy efficient CI/CD Pipieline

R26-SE-027

IT4010 – Progress Presentation 2

(PP2)

Agend
a

1) Problem & motivation

2) Solution overview

3) System Architecture & Integration

4) Commercialization

5) Demo

CURRENT PROGRESS AND NEXT PHASES

	PP1	
Release Component	Change Impact Analyzer, Selective Build Execution, Selective Test Execution done.	emi
Deploy Component	The energy profiler & deployment comparator Implemented	I decis witl
Operation	Future workload prediction	Jol


Energy

PROGRESS PRESENTATION 1 - COMMENTS

Student ID	Pa
IT22110084 (Release Component)	No
IT22149930 (Deployment Component)	No

The Real-World Problem

- Rapid growth of cloud-native systems and CI/CD pipelines creates green debt.
- DevOps pipelines prioritize speed over energy efficiency.
- Small code changes trigger full builds and tests.
- This causes resource waste, high energy usage, and increased carbon emissions.

Carbon Footprint

ICT sector produces ~2–4% of global CO_2 emissions.

Resource Waste

About 90% CPU time wasted during full rebuilds.

Economic Impact

35% annual increase in cloud costs.

Primary Stakeholders

- **DevOps/Platform engineers**
- **Software developers**
- **Site Reliability Engineers (SREs)**
- **Engineering Managers**

Our Solution Provides carbon-aware automation across every DevOps stage

RESEARCH GAP → OUR OBJECTIVE

RESEARCH GAP

Existing approaches optimize Green DevOps stages independently.

Disconnected carbon-aware decisions

OUR OBJECTIVE

Build one integrated Green DevOps framework that enables carbon-aware decision-making across all four stages.

Sustainability without sacrificing operational requirements

⚡ Delivery Velocity 📈 Performance ✓ SLA Compliance

SOLUTION OVERVIEW

An integrated Green DevOps framework embedding carbon-aware decisions across Release, Deploy, Operate & Monitor

- **Release** — Carbon-aware CI/CD
- **Deploy** — AI deployment scheduling
- **Operate** — predictive autoscaling
- **Monitor** — Anomaly detection & Sustainability Health Score

Non Functional Focus

reliability & SLA compliance

delivery performance

environmental sustainability

OUR ENVIRONMENT SETUP

Oracle Jump Server + Reverse SSH Tunnel Setup

1) ADMIN PC / CLIENT



Admin PC

```
ssh -p 2510 <user>@147.14.144.192
```

1) SSH to
147.14.144.192:2510



Internet

ORACLE CLOUD FREE TIER INSTANCE

Jump Server / Bastion Host

Static Public IP

Publicly Reachable from Internet



- **Port 22:** SSH service
- **Port 2510:** Exposed reverse tunnel endpoint

Internal Mapping (Reverse Tunnel)

2510 -> Ubuntu Laptop:22

External clients connect to 147.14.144.192:2510
Traffic is forwarded through the reverse SSH tunnel to the Lubuntu Laptop on port 22.

2) Reverse SSH tunnel from Lubuntu -> Oracle

3) HOME / LOCAL NETWORK



Lubuntu Laptop
Local Server

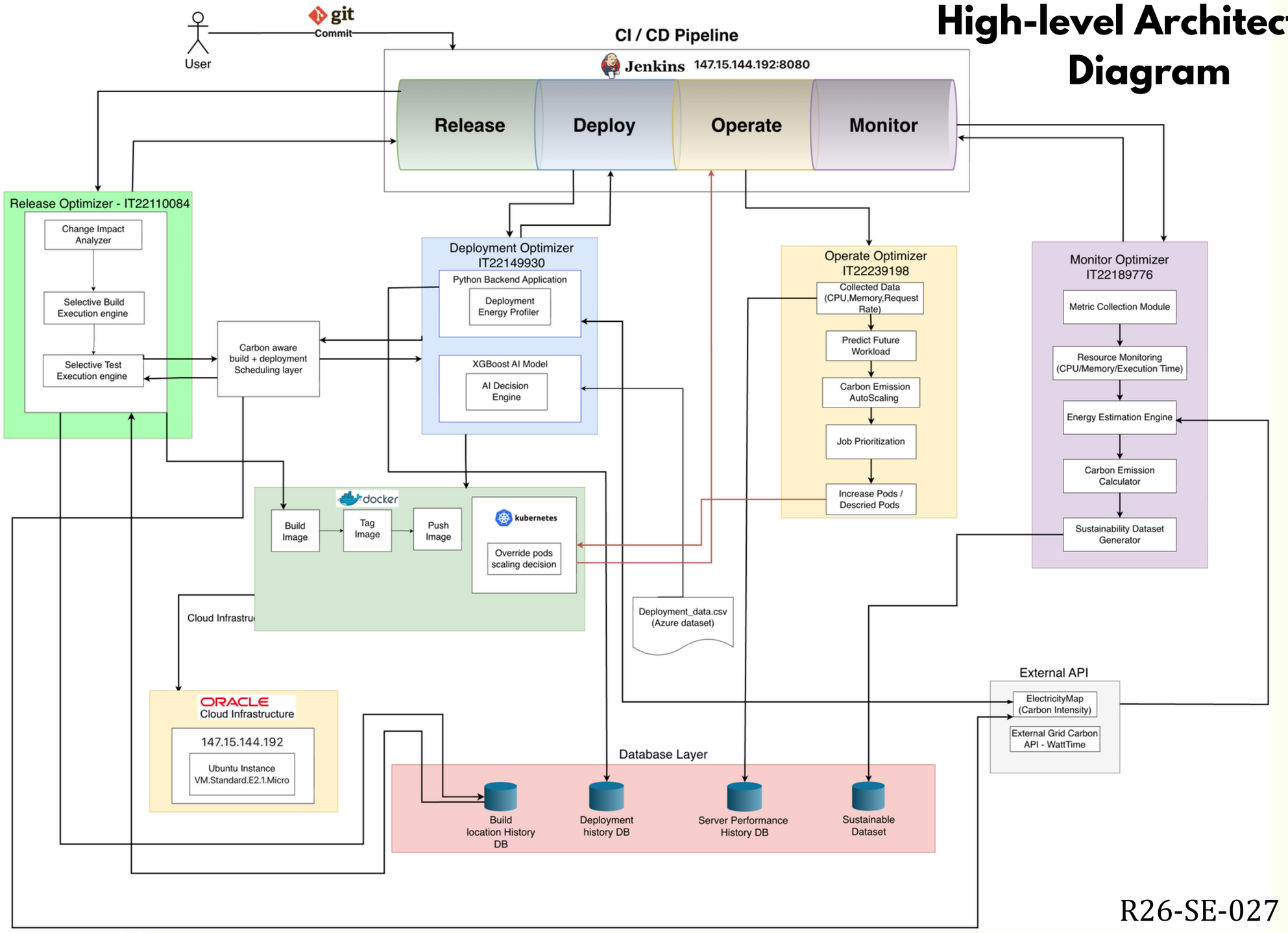
- Runs SSH server on port 22
- Behind NAT / no public IP
- Maintains persistent reverse SSH tunnel

```
autossh -N -R *:2510:localhost:22
opc@147.14.144.192
```

Admin PC → 147.14.144.192:2510 → Reverse SSH tunnel → Lubuntu Laptop:22

End-to-end: From Admin PC to Lubuntu Laptop via Oracle Jump Server

High-level Architecture Diagram



GREEN DEVOPS AS A COMMERCIAL PRODUCT

Modular subscriptions, complete-suite deployment and lifecycle support

1 PRODUCT OFFERING

RELEASE **Carbon-aware release scheduling**

Lower-carbon release windows • release recommendations

DEPLOY **Energy-efficient deployment**

Optimized deployment strategy • resource-aware rollout

OPERATE **Predictive Kubernetes operations**

LSTM prediction • carbon estimation • job priority • scaling decisions

MONITOR **Sustainability analytics dashboard**

Carbon, energy, cost and SLA visibility • requires connected CI/CD or infrastructure data

Individual module • selected bundle • complete 4-component product

2 SUBSCRIPTION & PRICING MODEL

FEATURE	STARTER USD 49/mo	PROFESSIONAL USD 129/mo	ENTERPRISE USD 249/mo*
Components	Any 1	Any 2–3	All 4
Dashboards	Core module view	Advanced cross-module	Executive + technical
Integration	Setup guide	Assisted integration	End-to-end integration
Maintenance	Updates only	Updates + health checks	Managed maintenance
Support	Business hours	Priority support	24/7 support
Best for	Pilot / small team	Growing DevOps team	Enterprise rollout

*Indicative academic pricing; final quotation depends on users, clusters, integrations, data volume and support scope.

3 COMMERCIAL DELIVERY

✓ Initial assessment & product configuration
✓ CI/CD, Kubernetes and monitoring integration
✓ Training, updates and ongoing maintenance
✓ SLA-based support, including 24/7 enterprise coverage

4 ADOPTION PLAN

1. Pilot one module with one team
2. Validate carbon, cloud cost and SLA results
3. Integrate selected modules
4. Roll out the complete suite across teams

5 CUSTOMER IMPACT

Lower cloud cost
Reduced carbon emissions
Efficient infrastructure use
Reliable and sustainable delivery

TARGET MARKET • Software companies • DevOps/platform teams • SRE teams • Kubernetes and CI/CD users

THANK YOU

Official Docs

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